

Homogeneous cases: $M \ddot{x}_{SI} + \gamma_{SI} \dot{x}_{SI} + k x_{SI} = 0$

define: $T_0 \equiv \sqrt{\frac{m}{k}}$ and $\gamma \equiv \frac{1}{2} \frac{\gamma_{SI}}{\sqrt{mk}}$

and then define $\tilde{t} \equiv \frac{t}{T_0}$

then:

$$\ddot{x}_{SI} = \frac{d^2}{dt^2} x(\tilde{t}) = \frac{1}{T_0^2} \ddot{x}$$

$$\dot{x}_{SI} = \frac{1}{T_0} \dot{x}$$

So

$$M \ddot{x}_{SI} + \gamma_{SI} \dot{x}_{SI} + k x_{SI} = 0$$

$$\frac{m}{T_0^2} \ddot{x} + \frac{\gamma_{SI}}{T_0} \dot{x} + k x = 0$$

$$\hookrightarrow k \ddot{x} + \frac{\gamma_{SI} \sqrt{k}}{\sqrt{m}} \dot{x} + k x = 0$$

multiply
by $\left(\frac{\sqrt{k}}{\sqrt{k}}\right) \left(\frac{2}{2}\right)$

$$k \ddot{x} + 2 \underbrace{\left(\frac{\gamma_{SI}}{2\sqrt{mk}}\right)}_{\gamma} k \dot{x} + k x = 0$$

thus, $k \ddot{x} + 2\gamma k \dot{x} + k x = 0$

k cancels out:

This leaves us with the unitless form

$$\ddot{x} + 2\gamma \dot{x} + x = 0$$

This has the three solutions

(i) Underdamped, $\gamma < 1$

$$x(t) = e^{-\gamma t} (A e^{i\omega_0 t} + B e^{-i\omega_0 t})$$

$$\omega_0 = \sqrt{1 - \gamma^2}$$

(ii) Overdamped, $\gamma > 1$

$$X(t) = A e^{\lambda_1 t} + B e^{\lambda_2 t}$$

$$\lambda_1 = -\gamma + \sqrt{\gamma^2 - 1}$$

$$\lambda_2 = -\gamma - \sqrt{\gamma^2 - 1}$$

limit Critical: $\gamma = 1$

$$X(t) = e^{-\gamma t} (A + Bt)$$

* A couple of notes: the exponent in front
 $e^{-\gamma t}$

has a negative for the unitless case but a (+) for the SI case. (As you see from the lecture notes).

* Use the above only for unitless ODE.
 Don't try and solve the SI unit one.

